Health Geography

Week 1 Discussion

Julia Lin

Department of Geography, UC Santa Barbara

Monday 5:00–5:50 PM
Thursday 9:00–9:50 AM

Where is your hometown?

Please submit your answer through Canvas before the end of section.

Format: *City or town, state/province, country*
(e.g., Santa Barbara, California, United States)



Today's goals

01 Section purpose & structure

02 Final project components

03 Neighborhood exposures vs. health outcomes

04 Research from everyday observations

05 Forming geographic research questions

06 Upcoming tasks: readings, quiz, field trip

Attendance reminder: Answer today's Question of the Week on Canvas before leaving.

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My doctoral research asks how individuals and households respond spatially to major life transitions and how those strategies shape health and well-being.

My research

- Population Health & Environmental Geography
- Spatial Demography & Mobility
- Geospatial Science & Technology
- Data Science & Computational Methods

Recent projects

- Food environments and subjective well-being
- Psychiatric service accessibility and crime
- Urban quietness
- Fertility decline and pet ownership in Taiwan

Office hours and communication

Office hours

Wednesdays, 3:30–5:30 PM

- Reserve a 15-minute appointment.
- Briefly describe what you would like to discuss.
- Choose an in-person or Zoom meeting.
- In person: Ellison 5723

Book an appointment

Zoom meeting link

Email

Please include:

- "GEOG 152" in the subject line
- Your name and discussion section
- A clear and specific question

I will respond as soon as reasonably possible during working hours. Please do not expect an immediate response during evenings or weekends.

If office hours do not work for you, email me and I will try to arrange another time.

What is section for?

1. Review & Apply Concepts

- Apply ideas from lectures and readings
- Practice interpreting research, data, and maps

2. Build the Final Project

- Learn the stages of the research process
- Develop the final paper and StoryMap
- Work through questions and problems together

Most weeks

1. Today's goals
2. Short concept review
3. One research skill or project step
4. Individual or small-group work
5. Upcoming class reminders

> Section is where you learn to conduct a real **health geography research project** from start to finish.

A few section expectations

- **Be present & engaged:** Attend your enrolled section and arrive on time.
- **Contribute to your team:** Actively participate and complete your share of group work.
- **Be respectful:** Focus critiques on evidence and treat all communities with respect.
- **Communicate proactively:** Ask for help before a small challenge becomes a large problem.

One-time schedule conflict:
Contact me before attending the other section.

Ongoing circumstances:
Contact Professor Cassels or me to discuss accommodations.

The research process

Research stage	What we will work on
Interest and motivation	What do you want to understand, and why does it matter?
Research question	What specifically are you asking?
Exposure and outcome	What neighborhood condition and health pattern will you examine?
Hypothesis and mechanism	What relationship do you expect, and why?
Data and methods	What evidence can help answer the question?
Results and interpretation	What patterns appear, what might explain them, and what are the limitations?
Communication	How will you present the research in a paper and StoryMap?

> If you keep up with the section activities, your project will develop gradually throughout the quarter.

Neighborhood research project

Investigate how one social or environmental neighborhood exposure may be related to health outcomes in two contrasting places within Santa Barbara County.

Geographic patterns

- How do neighborhood conditions and health vary across places?
- Why might these geographic differences exist?
- What evidence supports the proposed relationship?

Interpretation

- What other explanations might exist?
- What can and cannot be concluded?
- How can the findings be communicated to a broader audience?

Individual paper and group StoryMap

Individual research paper

- One neighborhood exposure (from a provided list)
- One unique health outcome (of your choice)
- Two contrasting local neighborhoods
- A research question and hypothesis
- Academic and publicly available evidence
- Maps, figures, observations, and photographs
- Interpretation and limitations

Group StoryMap and presentation

- One shared neighborhood exposure (from the list)
- The same two neighborhoods
- Different health outcomes (each member's choice)
- One combined geospatial story
- An ArcGIS StoryMap and oral presentation
- A peer assessment of group contributions

> Your individual findings contribute to the group's shared StoryMap.

Group formation in Week 3

During **Week 3,** you will identify:

- Neighborhood exposures you are interested in
- Health outcomes you may want to investigate
- Any special requirements or scheduling constraints

Professor Cassels and I will use these preferences to form balanced groups of approximately four students.

Within each group

- Members share one exposure.
- Members study the same two local neighborhoods.
- Each student studies a different health outcome.
- The group produces one combined StoryMap.

You do not need to choose your final exposure today.

Possible neighborhood exposures

Economic and social

- Social Vulnerability Index
- Median household income
- Poverty
- Unemployment
- Educational attainment
- Health insurance coverage
- Food insecurity

Housing and households

- Renter occupancy
- Overcrowding
- Single-parent households
- Disability
- Age distribution

Population and community

- Language spoken at home
- Latino population
- Foreign-born population
- Veteran population
- Another approved exposure with available data

> Your group will eventually select one shared exposure. We do not need to define or select every exposure today.

Possible health outcomes

Low birthweight	Obesity
Diabetes	Asthma
Cardiovascular disease	Smoking
Physical inactivity	Poor mental health
Sexually transmitted infections	Injury
Life expectancy	Self-rated health

Important

Each group member will investigate a **different health outcome** related to the shared exposure.

Before finalizing an outcome, we must confirm that appropriate data are available at a useful geographic scale.

An **exposure** is a social or environmental condition. A **health outcome** is a health status, behavior, condition, or experience.

Poverty and diabetes

STARTING INTEREST

Why do some communities experience a greater burden of chronic disease than others?

WHY IT MATTERS

Diabetes affects quality of life, health-care needs, and mortality. If neighborhood conditions contribute to unequal risk, individual behavior may not be the only appropriate focus of intervention.

EXPOSURE

Households below the poverty level

HEALTH OUTCOME

Estimated diabetes prevalence

Possible links between poverty and diabetes

EXPOSURE

Neighborhood poverty

→

POSSIBLE PATHWAYS

- Nutritious food access
- Physical-activity opportunities
- Chronic stress
- Transportation and health-care access
- Housing and working conditions

→

HEALTH OUTCOME

Diabetes prevalence

> A plausible pathway helps us develop a hypothesis. It does not prove causation.

Research question and hypothesis

> **Research question**
> How do poverty and estimated diabetes prevalence differ between two contrasting census tracts in Santa Barbara County, and what neighborhood conditions might help explain the observed patterns?

Hypothesis

The census tract with a higher percentage of households below the poverty level will also have a higher estimated prevalence of diabetes.

Important limitation

Finding that poverty and diabetes are both higher in one tract would not prove that poverty caused diabetes among particular residents.

Data and evidence

Neighborhood data

- Cottage Data2Go
- U.S. Census
- Maps and tables

Health evidence

- CDC PLACES
- Academic literature
- Publicly available estimates

Local context

- Neighborhood observations
- Photographs
- Alternative explanations

Open Cottage Data2Go

Today we will preview the geographic data available for your project.

The final StoryMap

- A motivating question
- Written explanations
- Interactive or static maps
- Data, charts, and tables
- Neighborhood photographs
- Individual health-outcome findings
- Interpretation and limitations
- Sources and credits

Examples

Data2Go NYC

Life-expectancy data story

ArcGIS StoryMap examples

Later in the quarter, we will learn how to use ArcGIS and build the group StoryMap.

Introductions and initial interests

Form a group of 3–4 students.

Introduce yourself

- Your name
- Your major or year
- One place-and-health issue that makes you curious

Your idea might come from

- Something you have noticed in everyday life
- A difference between two places
- A public-health or environmental concern
- A surprising pattern in the news
- Something you want to understand

5

MINUTES

Everyone should have a chance to share one idea.

You do not need to share personal health information.

Your group's initial idea

Open the group discussion slides

Edit only your group's assigned slide.

01 **We are curious about…**

02 **This matters because…**

03 **Possible neighborhood exposure**

04 **Possible health outcome**

05 **Possible pathway connecting them**

10

MINUTES

What do we want to find out?

Why might someone care?

How is this connected to health and place?

Your draft research question

> How might **[a neighborhood exposure]** be related to **[a specific health outcome]**?

Before you finish, check:

- Does the question arise from your original curiosity?
- Have you explained why it matters?
- Are the exposure and outcome identifiable?
- Does the question avoid assuming causation?
- Are all group members' names on the slide?

5

MINUTES

Write one complete question using your exposure and health outcome.

The Ghost Map: Follow the investigation

As you read, follow how an everyday health crisis gradually becomes a geographic investigation.

01

The problem

What is happening, and what makes the outbreak difficult to understand?

02

The explanations

What do different people believe is causing cholera? What assumptions shape their explanations?

03

The evidence

What observations, patterns, or clues might support or challenge those explanations?

> **Keep this question in mind:**
> What does paying attention to **where** people live, move, obtain water, and become ill make visible?

You do not need to memorize every person or event. Focus on how questions, explanations, and evidence develop.

Macintyre et al.: Opening the "black box" of place

The authors argue that it is not enough to conclude that "where you live matters." Researchers must explain **what about a place matters, for whom, and through what pathway.**

01

The problem

Why do the authors describe "place effects" as a **black box**? What is missing when researchers find an area effect but do not explain it?

02

What about place?

What examples do the authors give of **material infrastructure** and **collective social functioning** that could influence health?

03

How would we test it?

What must researchers specify about the pathway, population, spatial scale, and time period before studying a place effect?

> **Keep this question in mind:**
> How can we move from observing a geographic difference in health to proposing a **specific, testable explanation** for it?

Do not try to memorize every example. Focus on why the authors want researchers to replace vague "place effects" with specific mechanisms and testable hypotheses.

Before Wednesday's class

Readings for September 30

- Steven Johnson, *The Ghost Map*, pages 1–55
- Macintyre et al. (2002), "Place Effects on Health"

Reading quiz

Complete the quiz on Canvas before class.

Wednesday at 2:00 PM

Goleta Sanitary District

Choose one tour

- Monday, October 5 at 2:00 PM
- Tuesday, October 6 at 2:00 PM

The tour will last approximately 75 minutes.

Meeting location

Large flagpole at the Goleta Sanitary District
1 Moffett Place, Goleta, CA 93117

Before the tour

- Sign up for one tour.
- Read the guidelines on Canvas.
- Plan your transportation.
- Walk, bike, take the bus, or carpool.

Goleta Sanitary District

MTD Line 11

Final checklist

01 Submit the Question of the Week.

02 Confirm that your name is on the group slide.

03 Check the required readings and quiz.

04 Select a field-trip time.

05 Read the field-trip guidelines and plan transportation.

Today's main takeaway

A health geography project begins with **curiosity about differences across places**, and develops into a focused question about an **exposure**, a **health outcome**, and a **possible pathway** connecting them.

Questions?